Data Augmentation Based Cross-Lingual Multi-Speaker TTS using DL with Sentiment Analysis

LALITHA B.

Department of ECE, Sree Rama Engineering College 517501, Tirupati, India ; Email: lalitha.rohit@gmail.com

MADHURIMA V

Associate Professor, Department of Electronics and Communication Engineering, SV College of Engineering, Tirupati 517501, A.P, India; email: madhurimav16@gmail.com

NANDAKRISHNA CH

Department of Information Technology, Velagapudi Ramakrishna Siddhartha Engineering College, Vijayawada, India. Email: nkcherukuri@gmail.com

SATISH BABU JAMPANI

Department of CSE, Koneru Lakshmaiah Education Foundation, Guntur, India. Email: jampanisatishbabu@gmail.com

CHANDRA SEKHAR J. N. *

Department of EEE, SVUCE, Sri Venkateswara University 517502, India. Email: chandrasekhar.jn@svuniversity.edu.in

P. VENKAT REDDY

Department of CSE, Woxsen University, Hyderabad , Telangana State, India,; email: venkatreddy.patlilla@woxsen.edu.in

*Corresponding Author: JN Chandra Sekhar (Email: chandrasekhar.jn@svuniversity.edu.in)

Text to **S**peech (TTS) algorithms have made tremendous strides in recent years in terms of their ability to generate speech in a single language that sounds as natural as possible. However, because to a lack of available training data, the synthesis of speech from the same person in various languages continues to be difficult. It might be challenging to locate people who are proficient in numerous languages at a level equivalent to native speakers. Voice conversion is one method that may be used to create a polyglot corpus, which can then be used to solve this problem. This entails making use of a voice representation model that has been trained on 53 different languages through the application of hybrid deep learning in order to capture speaker-invariant qualities. In this study, we present a novel approach for the conversion of voices across different languages by employing Generated Adversarial Networks (GANs) to train a multilingual TTS system. The concept of individual likeness loss in order to address the special difficulty of maintaining one's individual speaking identity during the training process can be offered. This work is focused to provide the impression that voice data coming from a variety of languages and speakers was produced by the same individual, and one way to do so is by using this word. In order to determine the extent to which our model is useful, two experiments that compared it against benchmarks that made use of varying degrees of parameter sharing between languages are carried out. The purpose of these experiments was to evaluate the accuracy of pronunciation as well as the caliber of the synthetic voice during transitions between different languages.

Keywords: Cross Lingual Multi-Speaker Analysis, Text to Speech Analysis, Sentiment Analysis

1 INTRODUCTION

The application of approaches that make use of neural networks has allowed for great progress to be made in the area of Text to Speech (TTS) synthesis. When a Tibetan training corpus is utilized alongside other languages, Mandarin speech synthesis is unaffected. This is in contrast to previous cross-language frameworks including the Tibetan multilingual framework [1]. In addition, the hybrid BLSTM-based cross-lingual speech synthesis architecture [2] does not need the Tibetan monolingual architecture because it only needs sixty percent of the training corpus to synthesize a voice that is comparable to the Tibetan one. This suggests that a big resource language corpus may be used for voice synthesis of a low-resource language.

Enhancing speech across multiple languages is of paramount importance as it plays a vital role in facilitating effective communication, bolstering accessibility, and enabling diverse applications in our progressively multilingual and interconnected global

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landscape. This enhancement empowers individuals, enterprises, and institutions to transcend language barriers, broaden their audience reach, and ensure the utmost quality and usability of audio data across a spectrum of linguistic environments. Recent research has investigated many approaches to multilingual and multispeaker TTS synthesis. These approaches rely on unique databases of performing artists and require tweaking to accommodate additional voices. However, the expenses associated with these strategies are frequently rather significant. We present a novel solution to this problem in the form of a system that makes use of speakers who are proficient in more than one language to deliver high-quality speeches in a number of different languages.

Deep neural network (DNN)-based systems often require a large amount of data in order to be trained, which presents issues in environments with limited linguistic resources. Despite the fact that zero-shot multi-speaker DNN-based TTS systems have been created based on high-resource languages, their performance on unseen speakers continues to be below average. In this paper, we provide an end-to-end method that increases speaker similarity for both known and unknown speakers, with a special focus on accurate speech encoder modeling. This technique enhances speaker similarity for both known and unknown speakers.

Even when trained exclusively on monolingual datasets, recent developments in multilingual TTS systems have exhibited encouraging outcomes, which is cause for optimism. Cross-language voice synthesis, on the other hand, frequently results in the introduction of audible accents and is unable to match the quality of the speaker's original language. Due to the efficient exploitation of parallel processing throughout the training and/or inference processes, Transformer-based TTS models have demonstrated greater performance when compared to RNN-based models (for example, Tacotron citeshen2018natural). However, in a multi-speaker context with noisy data and diverse speakers, it is more difficult to learn the alignment between the text and the speech, which limits the application of Transformers in multi-speaker TTS.

The capacity to infer speech features from facial expressions has opened up new prospects for multi-speaker TTS technology, particularly in situations where a considerable volume of high-quality voice data as well as related transcriptions are available. However, the collecting of paired data using methods that require a significant amount of time continues to be a significant challenge for many research institutes. The phrase "multi-speaker speech synthesis" refers to the technique of synthesizing the voices of numerous persons from a single model. Deep neural networks (DNNs), despite the fact that various methods based on them have been devised, are prone to overfitting when the amount of training data available is restricted.

- WaveNet, an alternative to vocoders, has demonstrated extraordinary performance in the generation of very realistic synthetic speech in multi-speaker speech-to-text synthesis systems, notably in terms of naturalness and speaker similarity. This is due to WaveNet's ability to generate speech that is highly comparable to the natural speech of several speakers. However, in order for the WaveNet vocoder to work well, it is necessary to make accurate predictions of acoustic qualities, which are then employed as local condition parameters. The process of efficiently training this model is currently underway. The quality of synthetic speech suffers a major hit whenever the predicted acoustic qualities diverge from the actual ones. This causes a substantial decline in the quality of the produced speech.
- In the TSS model, text analysis is a very important component. The majority of traditional methods for analyzing text are predicated on rule-based approaches, which require a large time commitment for the purpose of collecting and learning these rules. Bigram analysis, trigram analysis, HMM-based analysis, and DNN-based analysis are just a few examples of the data-driven methodologies that have arisen as a result of the rapid developments in data mining technologies.
- When doing text analysis using the latter two approaches, the Festival system is typically utilized to carry out phoneme segmentation and annotation on the corpus. This is done so that the results may be better interpreted. This method requires many layers of analysis and considers a variety of factors, including the ones listed below:
- Phoneme level: This level takes into account the phonetic symbols that come before, after, or are located in close proximity to the present phoneme. It also takes into account how far along the syllable the current phoneme is located.
- Syllable level: At this level, the emphasis is placed on aspects such as syllable stress (previous, current, or next), the number of phonemes in the previous, current, or next syllable, the position of the current syllable within the word or phrase (forward or backward), the number of stressed syllables immediately preceding or following the current syllable within the phrase, the distance to the nearest stressed syllable (forward or
- Analysis at the word level takes into consideration the part of speech (POS) of the previous, current, or next word; the number of syllables in the previous, current, or next word; the position of the current word within the phrase (forward or backward); the content word (forward or backward) that is closest to the current word within the phrase; the distance to the nearest content word (forward or backward); and the POS of the previous, current, or next word.
- Phrase level: This level focuses on characteristics such as the number of syllables and words in the previous, current, or next phrase, as well as the location of the current phrase inside the sentence (forward or backward), and the prosodic annotation of the current phrase.
- At the highest level, the analysis takes into account the amount of syllables, words, or phrases that are contained in the sentence at hand.
- Because these several levels of analysis are taken into consideration, the TSS model is able to properly evaluate and comprehend the textual material, which in turn enables correct voice synthesis.

1.1 Preliminary Knowledge

TTS systems typically consist of three primary components: a text analysis front-end, an acoustic model, and a voice synthesizer. These components are educated independently from one another and need substantial topic expertise. Nevertheless, independent training may be a time-consuming process, and mistakes can compound as they are introduced into each component. The first step in the TTS synthesis process is an analysis of the text, and the second step is the generation of the speech waveform as mentioned in Figure.1. An approximation formula is used in order to convert the phonetic and prosodic information that was gleaned from the analysis into a waveform. In order to generate the correct speech output, the amplitude of each signal contained inside the speech waveform must first be measured. Depending on the circumstances, these speech waves might either take the shape of language or non-linguistic communication. The purpose of this investigation is to create a voice synthesizer that not only converts TTS but also stores the resulting file in the mp3 file format.

End-to-end speech synthesis technologies have become more popular in the industry as a solution to the problems that are related with the training of separate component parts and the potential accumulation of errors. These solutions include all of the components into a cohesive framework, which provides a number of benefits for TTS systems, including the following: (1) They are able to be trained using a large-scale dataset with minimum human annotation; (2) They do not require phoneme-level alignment; and (3) Errors do not compound since they rely on a singular model.

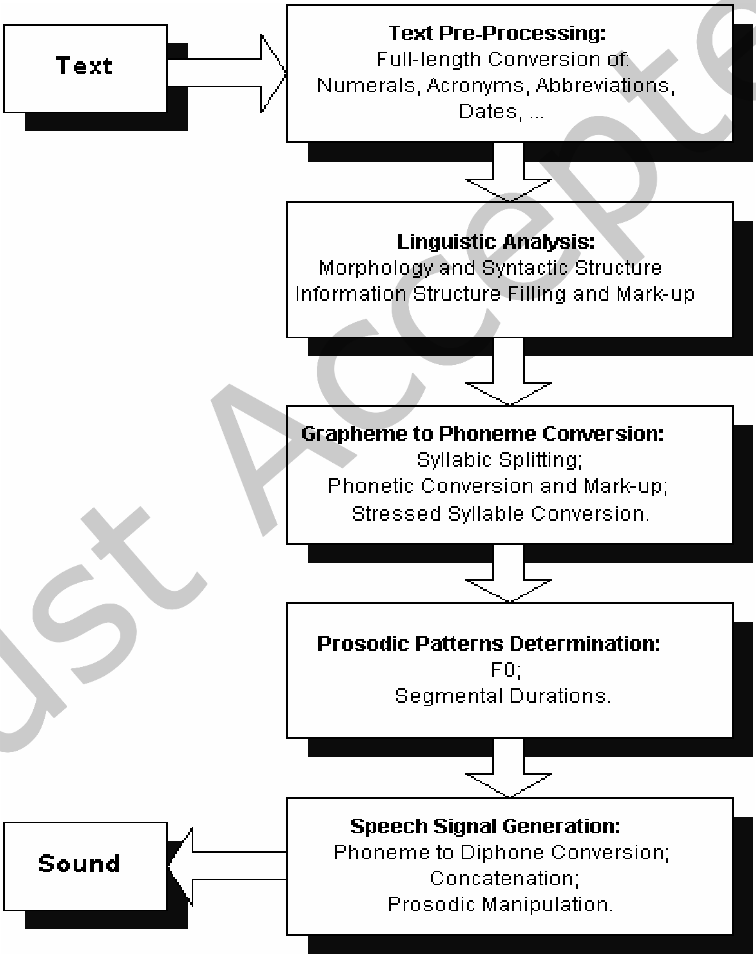


Fig.1. System Model

- The following are the stages that are often included in a TTS procedure:
- Text Input: To create speech in a particular language (for example, Punjabi), you must first provide the system with text written in that language. The text that is inputted might be as simple as a single word or as complex as an entire paragraph or collection of words. This input is subjected to further processing, which includes tokenization and the extraction of phonemes.

- Tokenization is the process of generating tokens by processing the text. This entails transforming the text into more manageable pieces known as tokens and deleting items such as numerals or special characters that are not desired. This preliminary phase of processing guarantees that the data are free of errors and prepared for the subsequent processing steps. Following this stage, the tokens that were produced are sent on to the subsequent procedure.
- Phonetic Synthesizer: The phonetic synthesizer is in charge of producing the tokens that are used in the game. The phonemes are extracted from each token, and then those phonemes are represented as location numbers within a data format. This operation is performed on each token. Each token is then transformed into a list of numbers, where each number represents the position of a phoneme in the list of phonemes included within the phoneme database. The token list is set to a predetermined size in order to preserve the consistent data form that is necessary for machine learning models. Empty items in the token list are replaced with the integer value '-1'. The model is then trained using this data, and its accuracy may be evaluated using any text that is entered.
- Formant Synthesis: This approach of generating voice from processed text is known as formant synthesis. This requires the utilization of formant frequencies as well as a particular strategy for converting these formant frequencies into the sound that is sought in Punjabi. The finished recording of the speech is saved in a separate file.
- Speech Output: This stage involves processing the speech that was created earlier in the procedure. The formant synthesizer model is used to evaluate the output of the speech generator from the stage before this one. In order to determine whether or not the speech output is accurate, the formant frequencies and any other attributes that have been extracted are compared to the database.

1.2 The Contributions Made to Research

Text is analyzed through the lens of sentiment in order to identify whether it has a good, negative, or neutral tone. This study is absolutely necessary for applications that are dependent on the views of the general population. In this study, the categorization of text based on emotion is investigated. A variety of label categorizations are shown for the text that is supplied as part of this investigation.

In the course of the study, we utilized a number of different cross-lingual transfer learning strategies in order to educate our low-resource TTS models. In the beginning, we just used a tiny amount of data that was already available in the targeted language and paired it with data that was already available in the target language. After that, both datasets to do additional adjustments to the model are used. Following the training of the algorithm with high-resource speech data, which is followed by training with target language data and enhanced target speech data, our studies show that this technique enables the system to generate extremely accurate speech synthesis in the target language. This strategy entails training the algorithm with target language data and augmented target speech data. It received an opinion score of 3.50 on average from native speakers.

It is essential to keep in mind that sentiment analysis models developed for a single language could not work very well when applied to data written in many languages and scripts. On the other hand, by making use of cross-lingual models that were educated on the training dataset, an average F1 score of 95.4% on the test dataset has been attained.

2 RELATED WORK

To analyze speaker attributes and generate synthesized speech in a variety of languages, we propose a neural end-to-end TTS system as part of this study. In our approach, we build a neural speaker modeling network that is uncoupled from linguistic and acoustic input during training. This illustrates that a voice synthesis network trained in English can be used to synthesize speech for Chinese speakers, and that a speech synthesis network trained in Chinese can be used to synthesize speech for English speakers. With just a few minutes of audio from a fresh speaker, our experimental findings suggest that the proposed model may reach acceptable levels of naturalness and similarity in speech synthesis for two languages [16]. Exploration of many options for generalizing the model to new speakers with less extensive speech data is possible.

As a solution to the challenge of adapting TTS systems to a wide variety of languages, it is suggested a speaker adaptation approach that combines domain adaptation approaches with a speaker consistency loss. In contrast to traditional methods, which rely on direct fine-tuning for adapting speakers within a single language, we use speaker verification in conjunction with domain adaptation to train a language-independent speaker encoder using multilingual data that includes both the source language and the language that will be used as the target. Then, we use the speaker embeddings generated by the speaker encoder to train a multilingual multi-speaker TTS model on the source language data. In order to fine-tune the source language TTS system utilizing target language data, we provide a speaker consistency loss. This loss seeks to increase the degree to which synthesized speech and the resulting speaker embeddings are highly correlated with one another. The speaker consistency loss is used to optimize this correlation. It can be shown that our proposed technique outperforms the gold standard process in terms of the spoken naturalness of the final translation by doing extensive tests on datasets that include both English and Japanese native speakers [17].

Detailed explanation on a whole learning-based architecture developed for cross-lingual Chinese and Tibetan voice synthesis is done. This approach allows for the simultaneous synthesis of Mandarin voices and Tibetan speech inside a single system. We do this by providing our acoustic models with both a large Mandarin multi-speaker dataset and a smaller Tibetan single-speaker corpus to train on. The system that provides context-based labels for phrases in both Chinese and Tibetan has been enhanced with the addition of a Tibetan text analyst to the textual analyzer. Our approach employs deep neural networks (DNNs), hybrid bidirectional LSTM (BLSTM), and hybrid long short-term memory (LSTM) to train acoustic models. Finally, we use a bilingual corpus of Mandarin Chinese and Tibetan to train long short-term memory (LSTMs), deep neural networks (DNNs), and back-propagation neural networks (BLNs) to construct synthetic voices. Using speaker adaptation approaches and data on Mandarin or Tibetan speech from a limited listener corpus, the models are fine-tuned. Finally, we use speaker-dependent modeling of Mandarin and Tibetan [18] to reproduce the sounds produced by native speakers.

By fixing the issue of lacking data in DNN-based TTS systems for low-resource languages, our study contributes to the advancement of cross-lingual TTS synthesis with multiple speakers. Problems with low-resource languages, such as the transmission of accents and the improper pronunciation of words, may be circumvented by the use of tone preservation methods and data augmentation, respectively. Applying these methods across a range of languages improves intelligibility, target accent ratings, and perceived similarities between speakers of various tongues.

It is proposed in this work that the Normalizing Flows (NF) method, which is used in neural vocoding, be significantly improved upon. To be more specific, we look at the possibility of enhancing expressive voice vocoding by using a modified version of the Parallel WaveNet (PW). Instead of the PW's affine transformation, we propose using invertible non-affine functions for the transformation due to their greater expressiveness. The enhanced PW allows for more expressive waveform reconstruction and TTS results. We test our model on a broad range of speech patterns utilizing data from a number of speakers who each speak a different language in order to determine how well it performs. The results of the waveform reconstruction tests suggest that the model we have presented reduces the discrepancy between the original PW and the recorded waveforms by 10% in terms of the naturalness and signal quality. Furthermore, our model achieves a reduction of over 60% when compared to existing neural vocoding systems presently regarded to be state-of-the-art. When measured on a set created for the express aim of giving feedback, objective measures like as L2 Spectral Distance and Cross-Entropy show gains of 3% and 6%, respectively, over the affine PW. In addition, we extend the approach introduced in the PW study to any non-affine invertible and differentiable function through the distillation of probability densities. [19].

Given the resource constraints of cross-lingual TTS synthesis, we describe a technique that can synthesize speech in various languages using distinct voices. We zero down on synthesizing English and Mandarin voices using limited training data, including only 15 minutes of a female speaker's Mandarin speech. Common problems in low-resource languages, such as mispronunciation and accent transfer, are addressed by our method. We suggest solutions, including tone preservation and data augmentation, to cope with these difficulties. All of these methods enhance intelligibility, target accent ratings, and cross-lingual speaker similarity ratings when applied to a strong multilingual baseline [20].

Authors describe a method that uses tone/stress embeddings as an extension of language embeddings to add tone and stress into the synthesis process. The proposed system can produce synthetic speech with a variety of accents by changing the tone/stress embedding input. To make it simpler to incorporate new speakers into an online application scenario, authors condition a Tacotron-based synthesizer on speaker embeddings generated from a pre-trained x-vector voice encoder using transfer learning. Authors present a common phoneme set to promote further phoneme sharing beyond the International Phonetic Alphabet (IPA). Our Mean Opinion Score (MOS) ratings confirm the intelligibility and fluency of native speakers of every language. Authors also find that L2-norm normalizing and ZCA-whitening on x-vectors significantly improve the system's dependability and audio quality. WaveNet models may be trained on any of the three languages supported by our system, and authors discover that they can produce high-quality speech in the other two languages[21].

In order to increase the similarity of speech for unidentified speakers, this work presents SC-GlowTTS, a zero-shot multi-speaker speech synthesis system. In this paper, authors introduce the idea of zero-shot flow-based decoding and offer an outline of the system's architecture. The gated convolution encoder, the transformer-based encoder, and the dilated residue convolution encoder are only few of the text encoding algorithms authors investigate. Furthermore, authors show how the TTS algorithms' suggested GAN-based vocoder, which is specifically designed for spectrograms, may greatly improve the similarity and quality of voice for new speakers. State-of-the-art outcomes in terms of sound quality and speaker similarity [22] are achieved by our system with just 11 speakers.

This research aims to improve the efficiency of zero-shot speaker adaption for unseen speakers as a means of overcoming the data shortage barrier in DNN-based TTS for low-resource languages. To combat this problem, authors provide a unique multi-stage transfer learning strategy that employs a single-speaker TTS system equipped with a d-vector speaking encoder trained on a high-resource language as the starting domain. In addition, authors offer a TTS model that makes use of explicit style direction from target speech to improve zero-shot sentence adaption, as well as speaker representations loss at the phrase level during TTS training. Our study use open-source speech datasets to show how our training strategy efficiently prepares TTS models for languages with few accessible training materials. Our models, as compared to those trained using more traditional approaches, generate more understandable and lifelike speech. Our proposed style encoder networks, in conjunction with the speaker reconstruction loss, also show substantial improvements in speaker similarity over the baseline model in the zero-shot speech adaption test. When applied to the unknown speakers test set, our proposed TTS model and training approach increase the cosine resemblance between native and non-native speakers by 0.468 and 0.279, respectively [23].

A speech encoder utilizing the best-of-breed ECAPA-TDNN models from the speaker verification challenge, a synthesizer based on the FastSpeech2 model, and a vocoder using HiFi-GAN are the three parts of the proposed architecture that have been trained separately. In comparison to existing voice encoder models, our method achieves higher levels of speaker similarity. Authors adopt and evaluate multiple automated Mean Opinion Score (MOS) assessment methodologies based on deep learning, a first in this area [24], to reliably measure the quality of the synthesized speech.

Using a fine-grained style encoder to improve voice quality and allow cross-speaker style transfer to correct inauthentic accents, authors offer M3, a multi-speaker, multi-style, multi-language speech synthesis system. Using a speaker-conditioned variational encoder and adversarial speaker training with a gradient reversal layer, authors are able to block timbre information from entering the style encoder. To further link the textual output of each speaker with their unique extracted style vectors, authors create a Mixture Density Network (MDN). The inference process allows for the transfer of any speaker's style type into the target language, allowing for the production of speech that is both idiomatic and free of non-native pronunciations. Our suggested method routinely achieves a 4.0 [25] on the MOS-speech-naturalness measure, which is much higher than the performance of the baseline system.

In this paper, authors introduce MultiSpeech, a multi-speaker Transformer-based TTS system with a focus on synchronizing text and sound. To avoid training and inferencing twice, authors use two layers of normalization on the phonetic encoding in the encoding phase and a pre-net obstacle in the decoding phase. Authors show the efficacy of MultiSpeech via tests using the VCTK and LibriTTS multi-speaker datasets. Compared to naive Transformer-based TTS, our system's ability to synthesis multi-speaker audio is greatly improved by training a FastSpeech model on top of the MultiSpeech model [26].

This research presents Face2Speech, a new kind of speech synthesis system that makes use of facial image predictions to create realistic speech. The system includes a voice encoder, a text-to-speech (TTS) module with several voices, and a face encoder. While the voice encoder creates an individual embedding vector for each speaker, the face encoder's captured facial picture is used in the multi-speaker TTS module's encoding process. Synthetic speech created from a face-derived encoding vector is equivalent to that generated from a speech-derived embedding vector, as shown by our matching and naturalness assessments [27].

Authors present a semi-supervised learning strategy for multi-speaker TTS. The proposed encoder-decoder system with discrete speech representation allows us to train a multi-speaker TTS model using raw audio alone, eliminating the requirement for transcriptions. A single hour of matched speech data from numerous speakers or a single person is sufficient for the suggested model to create understandable speech in a variety of voices, as shown by experimental findings. Even with noisy paired voice data, authors find that the suggested semi-supervised learning strategy still benefits the model. In addition, authors show how speaker characteristics affect semi-supervised TTS using paired data, which is a major focus of our study. [28].

As a solution, authors offer a framework for multi-speaker sound synthesis that makes use of deep Gaussian processes (DGPs), a complex structure of Bayesian kernel regressions that is resistant to overfitting. In our method, authors use speaker codes to provide information about the speakers to time and sound models. The potential of deep Gaussian process latent variable models (DGPLVMs) is also investigated. Our approach concurrently learns the representation of each speaker along with other model parameters to accurately capture speaker similarities and differences. Experiments were run with both balanced data from all speakers (speaker-balanced) and unbalanced data from select speakers (speaker-imbalanced) to see which strategy performed better. Both DGP and DGPLVM are shown to perform better than a standard DNN in synthesizing multi-speaker speech in a speaker-balanced situation, according to the assessment results. Authors also find that DGPLVM performs better than DGP [29] in the speaker-imbalanced situation.

Authors propose novel WaveNet-based multi-speaker sound synthesis frameworks that use a single conditional generative adversarial network (GAN) or its version, the Wasserstein One GAN with gradient penalty (WGAN-GP), to overcome this difficulty. WaveNet's acoustic model takes inputs from the GAN generator and uses them to calibrate local condition parameters. Authors add the discretized mixture of logistics (DML) cost from a well-trained WaveNet to the standard mean-squared-error (MSE) and adversarial-loss (ADL) components of GAN frameworks. Models trained with WGAN-GP with back-propagated DML losses show better accuracy and speaker similarity in subjective testing [30].

The results of pertaining to augmenting data for automatic speech recognition (ASR) systems, particularly in scenarios where resources are limited or moderate experiments reveal that our approach effectively narrows the gap between ASR models trained with synthesized speech and those trained with human speech. This achievement is noteworthy when compared to other methodologies that involve the use of multiple speakers for similar purposes. Moreover, our findings indicate the feasibility of achieving promising ASR training outcomes through our data augmentation method, even when employing only a single real speaker from the target language [31].

3 PROPOSED WORK

3.1 System Model

Deep learning has gained popularity in various natural language processing (NLP) applications, including sentiment analysis. However, these methods typically rely on large annotated corpora, which can be challenging to obtain, especially for languages with limited resources. In order to bridge this knowledge gap and address the scarcity of resources in sentiment analysis, we propose a novel deep multi-task multi-lingual approach. Our system leverages cross-lingual word embeddings to effectively align different languages in a shared semantic space. This enables the seamless exchange of information across languages, facilitating more accurate sentiment analysis even in resource-constrained environments..

3.2 Cross Lingual Word Embedding TTS

Most cross-lingual embedding models either utilize monolingual training models directly or extend existing multilingual word embedding methods to handle cross-lingual contexts. As a result, the foundational techniques used for monolingual word embeddings can be adapted to their respective cross-lingual counterparts. Recent research on cross-lingual anchoring models has revealed that the effectiveness of these models depends more on the choice of multilingual signals rather than the architecture itself.

Cross-lingual procedures can be categorized along two dimensions: comparability and alignment. Comparability refers to the availability of parallel data sources, such as literal translations or equivalent data across languages. Alignment level, on the other hand, is related to the level of supervision provided for bilingual data. Various types of alignment can be performed on bilingual data, including word and phrase alignment similar to machine translation, as well as document alignment.

In our study, we adopt a word-level mapping approach using parallel data to establish a bilingual setting. Furthermore, we aim to maximize the similarity between the source and target word embeddings while imposing the constraint of orthogonality. This constraint ensures that the target-to-source translations remain orthogonal. By employing this method, the computation process is facilitated and streamlined.

3.3 GAN Model for Data Augmentation

To address the issue of an unbalanced dataset, down sampling is performed to ensure a well-balanced training dataset. This step was taken to improve the model's performance, as the initial dataset distribution was unfavorable. It is expected that the Cross Lingual model will outperform other models, particularly in both low-resource and high-resource languages. By training the GAN Model with a Translation Language Modelling approach, the model can learn common representations across languages, enhancing its ability to perform well.

Data Augmentation (DA) Technique is a valuable approach that allows us to artificially increase the size of the training data by generating various versions of the existing datasets without the need for additional data collection. However, when applying data augmentation in Natural Language Processing (NLP), particularly for language-related tasks, it poses unique challenges compared to computer vision. Unlike images, where it is relatively easy to create augmented versions, language data augmentation is more complex due to the intricacies and context sensitivity of natural language.

In NLP, we need to be cautious when replacing words with synonyms, as doing so may alter the context and meaning of the text. Therefore, direct word replacement is not always a suitable strategy. Nonetheless, data augmentation techniques are still valuable in NLP, as they help address issues of data scarcity and lack of diversity, ultimately improving the performance of classification tasks.

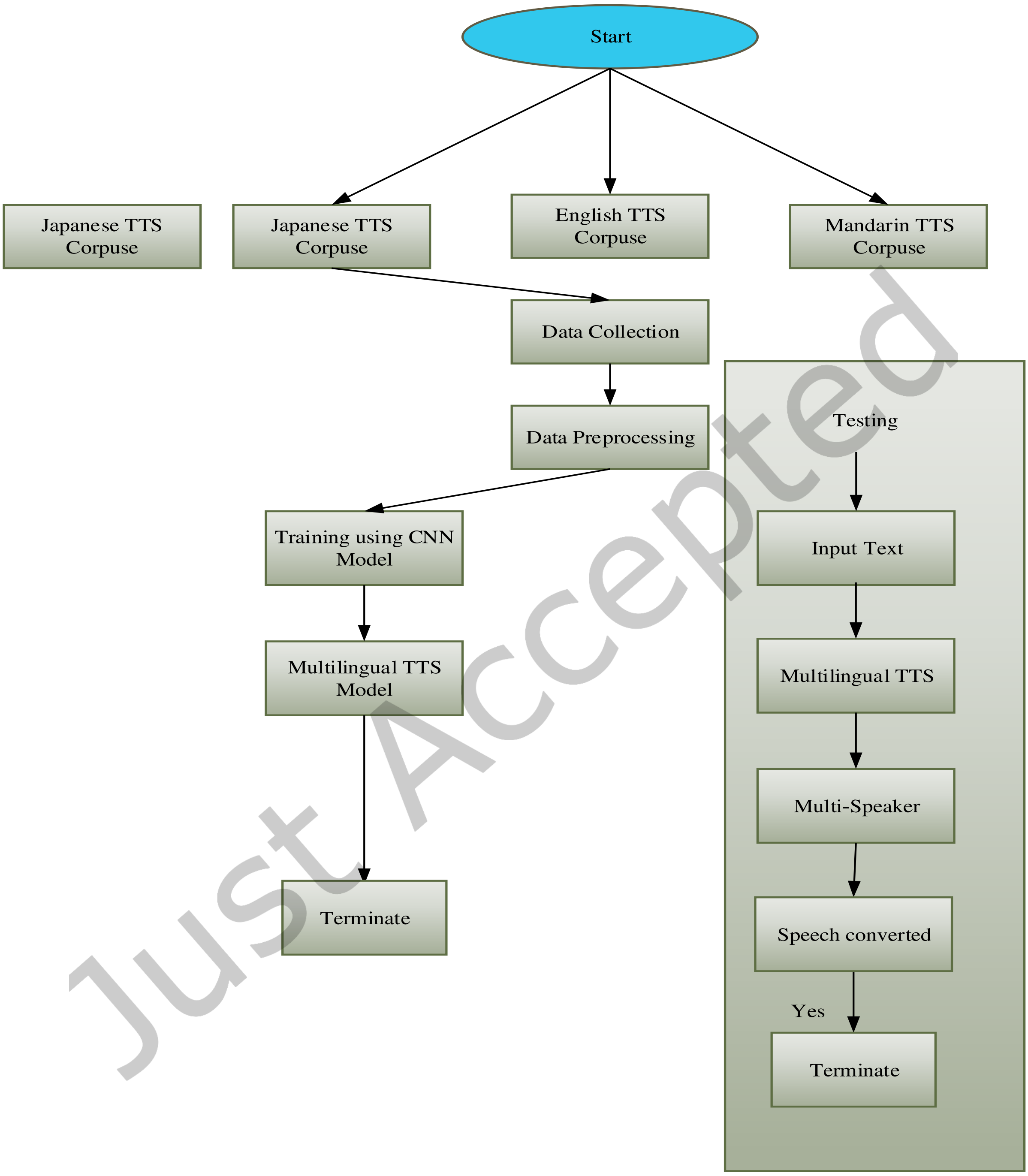


Fig.2. Flowchart for Proposed Method

Data augmentation is a technique that increases the size of the training data, leading to improved model performance. Having more data generally results in better model performance. However, it is important to strike a balance with the augmented data's distribution. The augmented data should neither be too similar nor too different from the original data, as this can cause overfitting or poor model performance. Effective data augmentation approaches aim for a balanced distribution. There are several benefits of data augmentation

- **Cost reduction:** Data augmentation helps reduce the costs associated with collecting and labeling large amounts of data.

- **Improved prediction accuracy:** By increasing the size of the training data, data augmentation improves the model's ability to learn and make accurate predictions.
- **Prevention of data scarcity:** Data augmentation addresses the issue of insufficient data, ensuring that the model has enough varied examples to learn from.
- **Reduction of data overfitting:** By creating variability in the data, data augmentation helps prevent the model from overfitting to the training set, leading to better generalization to unseen data.
- **Enhanced generalization ability:** Data augmentation increases the model's ability to generalize its learned patterns to new and unseen data.
- **Addressing data imbalance:** Data augmentation techniques can be used to alleviate the problem of imbalanced classes in classification tasks, ensuring that the model is trained on diverse examples from each class.

The model in question utilizes a 2048 encoder, pooling layers, 12 hidden layers, and a dropout rate of 0.1. Its maximum supported vocabulary size is 30145. Since the dataset consists of code-mixed Tamil and English text, the model incorporates language embedding to handle both languages. Both the training and validation datasets were used to develop the model

To condition a mel-spectrogram generating network, the model utilizes the speech signal of the target speaker. This is achieved by extracting acoustic properties using the speaker encoder. The structure of the speaker encoder is inspired by GAN-TTS, which captures universal speech features irrespective of speaker or language. The convolution layers in this model use filters with sizes of 64, 128, 256, and 512. The Generation Block processes the preprocessed raw audio as input. Figure 2 illustrates the structure and flow of the speaker encoder within the model.

. To achieve temporal smoothing of activations over an extended duration, a temporal average layer is employed. The pooling layer extracts significant features, followed by an affine transformation. Through the affine transformation layer, the utterance-level representation is projected into a 256-dimensional embedding. To create a speaker embedding that encompasses both the speaker's characteristics and language nuances from the latent space, the output undergoes regularization using length normalization. This process ensures that the speaker embedding captures the desired traits while accounting for variations in utterance length.

Let's first make a brief introduction to their structure that consists of two parts.

A data generator that is capable of learning to provide credible data. The generator component of the GAN architecture receives as input a random vector of a predetermined length, and its goal is to learn how to create samples that have distributions that are similar to those of the original dataset. After that, the samples that were created are sent through the discriminator's training process and utilized as examples of what not to do. The generator's goal is to trick the discriminator into thinking that the simulated instances are real ones by producing realistic samples that are accurate representations of the primary dataset.

A discriminator that is capable of learning to tell the difference between the generator's synthetic data and actual data. The discriminator part of the GAN system accepts a sample as its input and determines whether or not it is "real" (meaning that it was taken from the primary dataset) or "fake" (meaning that it was created by the generator). The discriminator is responsible for determining which samples are authentic and which ones are fraudulent, and providing input to the generator. If the generator produces samples that are impossible or unrealistic, it will be penalized by this feature, and those examples will be labeled as "fake." This adversarial process drives the generator to increase its ability to create samples that closely match the genuine data. The goal of this process is to fool the discriminator and limit the discriminator's capacity to differentiate between real and false samples.

- The generator wants to minimize $\log(1 - \mathrm{D}(\mathrm{G}(\mathrm{z})))$ where z is the element of randomness that is fed into the generator. The discriminator is fooled into thinking that the bogus samples are legitimate since the generator plays down the significance of this phrase.
- The discriminator wants to maximize $\log\left(\mathbf{D}(\mathbf{x_d})\right) + \log(\mathbf{1} - \mathbf{D}(\mathbf{G}(\mathbf{z})))$ where $\mathbf{x_d}$ These are representative examples taken from the initial dataset. This phrase refers to the chance of correctly labeling both the genuine examples and the samples that were generated by the algorithm.

The use of Generative Adversarial Networks (GANs) offers several advantages, including the ability to generate high-quality data and their versatility in addressing various problems. However, GANs also come with certain disadvantages and challenges. For instance, generating discrete data such as text can be challenging for GANs. GANs require extensive training and often demand significant computational resources. Similar to other Artificial Neural Networks (ANNs), GAN models can suffer from instability or overfitting issues.

In our GAN model, we employ the following parameters:

The activation function is Leaky ReLU, with a negative slope of 0.2.

- There will be five of each item in each batch.

- The learning rate is now set at 0.0002 percent per second.
- A dropout algorithm is used in the generator, and its probability is set at 0.3.
- The loss function is represented here by the binary cross-entropy.
- The Adam algorithm is being used as the optimizer at this time.
- Our design does not include any layers that are used for convolution.

In the event that the generator has many layers, those layers are arranged with the larger ones coming first, but in the discriminator, the layers are arranged with the smaller ones coming first. Dropout is a widely employed regularization technique within neural networks, particularly in the realm of deep learning, aimed at mitigating overfitting. Its core concept revolves around the random deactivation, or "dropout," of a specific proportion of neurons during each training iteration. This strategic process bolsters model robustness by reducing its dependence on specific neurons, ultimately enhancing its ability to generalize to previously unseen data.

The Adam algorithm, an abbreviation for Adaptive Moment Estimation, stands as a widely employed optimization technique frequently employed in the training of both machine learning and deep learning models. Distinguished for its efficiency, speed, and commendable performance across a diverse spectrum of tasks, Adam amalgamates the strengths of two other optimization algorithms, namely AdaGrad and RMSprop.

Before fine-tuning the entire model using triplet loss, which maps sound into a feature space where distances reflect speaker similarity, we initially train the speaker anchoring network independently on a speaker authentication task using softmax loss. The triplet loss requires three examples: an anchor, a positive example, and a negative example. During training, the model utilizes pairs of speaker embeddings to increase the cosine similarities between embeddings from the same speaker and decrease the cosine similarities between embeddings from different speakers. In essence, this process helps the model learn to distinguish between speakers based on their embeddings.

$$Loss = \sum_{i=1}^{N} \left[\|f_i^a - f_i^p\|_2^2 - \|f_i^a - f_i^n\|_2^2 + \alpha \right]_+ \quad (1)$$

where f_i^a, f_i^p and f_i^p are the embedded anchor, positive, and negative speakers, $\alpha \geq 0$ is a constant.

Through training the model on data from multiple languages, we enable a two-step cross-lingual reasoning process. Initially, the model generates the voice of a native speaker in the target language, followed by converting that voice to match the target speaker. The evaluation results indicate that, on average, our technique enhances the intelligibility of the generated output compared to the baseline. Furthermore, it suggests that our approach might also improve the pronunciation of cross-lingual output. Notably, the same model demonstrates its efficacy even in low-resource scenarios, requiring only a single training procedure. This is achieved despite the inclusion of speakers in the training set with an average speech duration of 7.5 minutes.

3.4 CNN-Bi_LSTM Model for TTS

The key component of our model that is responsible for categorization is known as an attention-based bidirectional long short-term memory (Bi-LSTM). Though the Convolutional Neural Network (CNN) may minimize the number of characteristics that are input in order to make more accurate predictions, it does not address the connection that each word has with the final classification in the same way. In order to make the most of the capabilities offered by both the models, the authors of this work make use of Bi-LSTM to encode long-distance word relationships in a more efficient manner.

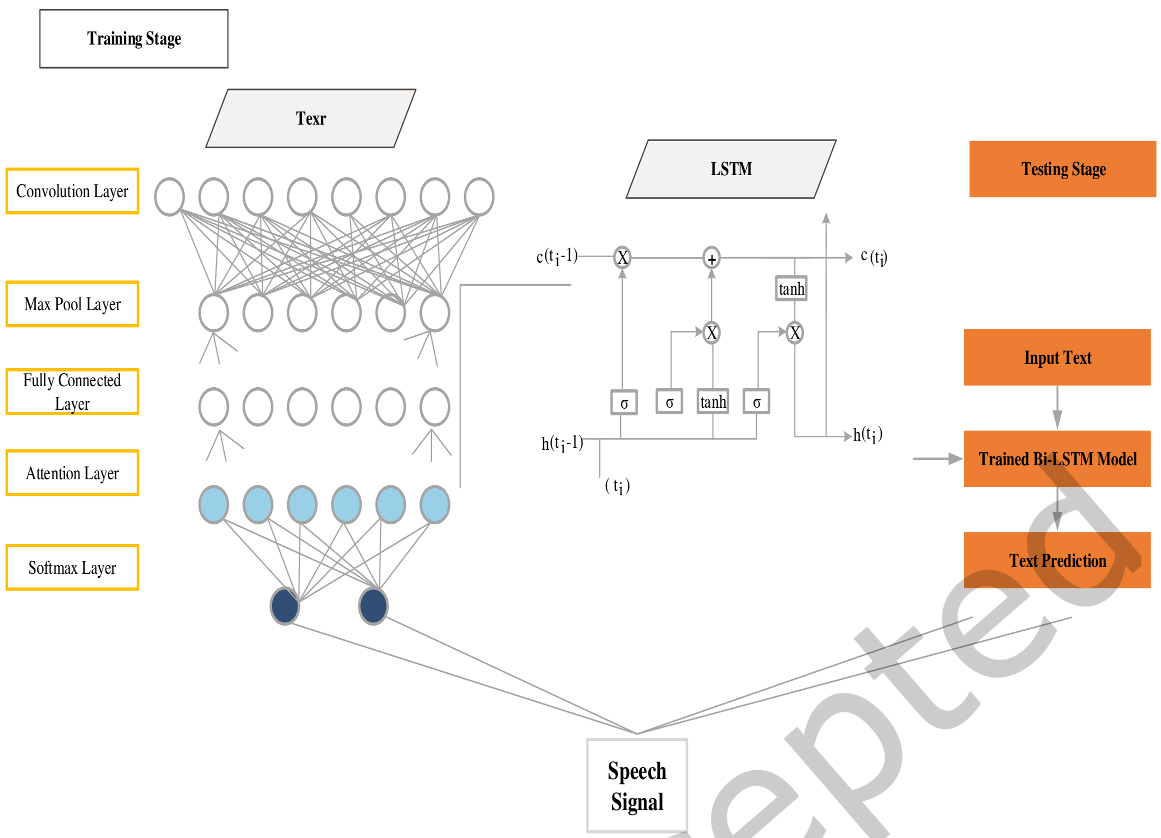


Fig.3. TTS using CNN with Bi-LSTM

Figure 4 is an illustration of the Bi-LSTM method, which depicts how it takes the features created by the CNN stage as input and uses those features to extract features from the final hidden layer. The Bi-LSTM is able to generate two separate textual representations due to the fact that it takes into account both the prior and following contextual information. After having been acquired through CNN_ij, the aforementioned characteristics are subsequently introduced into a Bi-LSTM model in order to provide a sequence representation. The resulting feature representation is then put through an attention layer, which identifies the strongly associated characteristics that will be used for the classification in its entirety.

Take, for instance, the line "Though he is sick, he is happy," which has to be categorized as having a positive mood. "Though he is sick, he is happy" According to the attention technique, the word "happy" is given more weight than the word "sick," showing that the feeling of "happy" is unaffected by the incidence of sickness. The accuracy of predictions may be greatly improved with this attention strategy, and the amount of learnable weights that are necessary for prediction can be reduced. In the context of the model that we have developed, we make use of the Bahdanau attention approach, with attention scores being calculated as follows:

$$\mathrm{score}(querry, key) = V^{\mathrm{T}} \tanh\left(W_{1key} + W_{2query}\right) \quad (2)$$

4 RESULTS & DISCUSSION

In this part of the article, we will discuss the experiments that were carried out in order to evaluate how well the suggested model works. The outcomes of the experiments are analyzed and compared to the most advanced deep learning models that are currently available. Both the number of epochs and the size of the data set were used as independent variables in two separate trials. The following text describes the experimental conditions, data utilized in the experiment, and assessment techniques.

The following is a concise summary of the datasets that were employed in the experiment that we carried out. Following this, the proposed model is disassembled into its component elements, and a variety of different training strategies are presented. At long last, the outcome of the subjective evaluation has been disclosed.

In order to produce our dataset, we collect four speech corpora in their respective languages from both internal and external sources. The information shown in Table 1 reveals that the dataset contains multi-speaker transcripts in the languages of Mandarin Chinese (ZH), English (EN), Korean (KO), Japanese (JA)and . The frequency of audio files is resampled to 22.05 kHz after the process. Because we have a grapheme-to-phoneme converter in-house, we are able to translate speech transcripts into a string of phonemes. A validation set that excludes 5% of the total phrases is considered complete.

Table 1: Language-specific information about our dataset

Language	EN	KO	JA	ZH	Total
Quantity of speakers	162	28	111	175	473
Length	72.4hr	43.5hr	27.5hr	60.0hr	203.4hr

Additionally, we performed data augmentation on the selected 30 minutes of multi-speaker speech data, generating approximately 15 hours of augmented target language data. This augmentation involved altering the pitch and speed of the original speech. The pitch shift was conducted in increments of 0.5 semitones, ranging from -2.5 to 2.5 semitones. The speed ratio of the augmented speech compared to the original speech ranged from 0.7 to 1.55 times the original speed, in steps of 0.05. This resulted in 27 variations of each target language utterance.

During the training process, we utilized two NVIDIA A100 GPUs and trained our framework for 200 iterations. The batch size for mixed-precision training was set to 64. The speaker classifying loss in DAT [20] was weighted according to a sequence of scale factors, following the prescribed method.

$$\lambda = \frac{2}{1+\exp(-10 \cdot p)} - 1 \qquad (3)$$

W

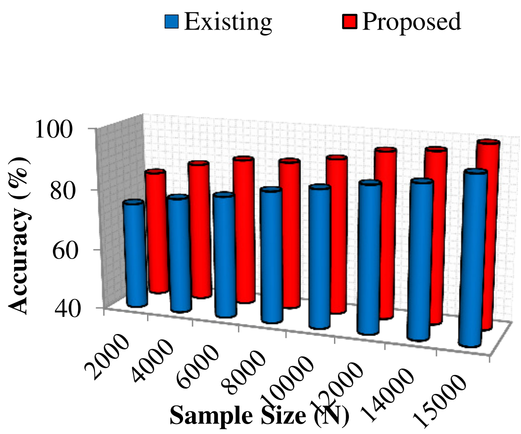
The factor p, which ranges from 0 to 1 in a linear manner, is used to determine the loss scale factor during training. Initially set to 0, it gradually increases as the number of training iterations progresses. This schedule helps to mitigate DAT (Discrete Anomaly Training) in the early stages of training and improves the quality of the generated outputs.

To demonstrate the high-quality statements generated by GAN-TTS during cross-lingual inference, we compare it to an official open-sourced version of a separate multilingual TTS model. The meta-education model is trained with a batch size of 64 for up to 100 iterations, using the Griffin-Lim vocoding method included in the official implementation.

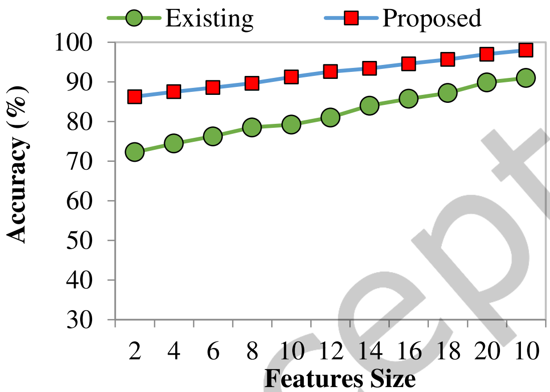
We evaluate GAN-TTS by comparing it to models where a single tweak has been removed. Specifically, we remove the use of DAT, include speaker regularization loss, and switch from SDP (Soft Dynamic Programming) to DDP (Dynamic Duration Prediction). During cross-lingual inference, we input the speaker embedding into the duration predictor after removing the speaker regularization loss. For the international TTS model to function, we do not make any adjustments to the language embedding.

To assess the resemblance of the speaker and the naturalness of the utterances, we use mean opinion scores (MOS) with 95% confidence intervals. The MOS assessment employs an absolute subcategory rating system, where performances are rated on a scale from 1 to 5 with 1-point increments. We collect opinion ratings on English speech from native English speakers, as it is challenging to find native experts for evaluating non-English target languages. Each individual utterance and speech pair is evaluated by a total of five raters. Speaker similarity is determined by comparing synthetic speech to actual speech by the same speaker from the validation set. To establish ground truth, two speeches by the same speaker from the validation set are compared by human raters to assess their similarity. Thirty phrases from the English validation set are randomly chosen as evaluation speech samples. Additionally, we test the intra-lingual and cross-lingual synthesis ability by selecting five male and five female speakers from each of the four languages, resulting in over 1200 different combinations of voices and phrases being synthesized.

Based on the performance metrics of the model, sentiment analysis of code-mixed Tamil-English text still has room for improvement. If the output is affected by an imbalanced dataset, up-sampling procedures can be employed to rectify it instead of down-sampling. The misclassification of text using the test dataset may be attributed to the loss of certain critical characteristics due to down-sampling. Considering that the retrieved YouTube postings may contain symbols and abbreviations, preprocessing the data is as crucial as up-sampling the system.

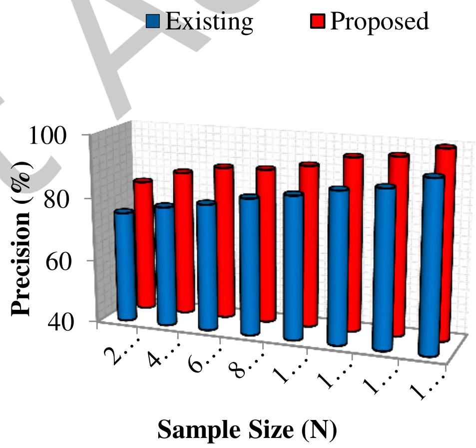


(a)

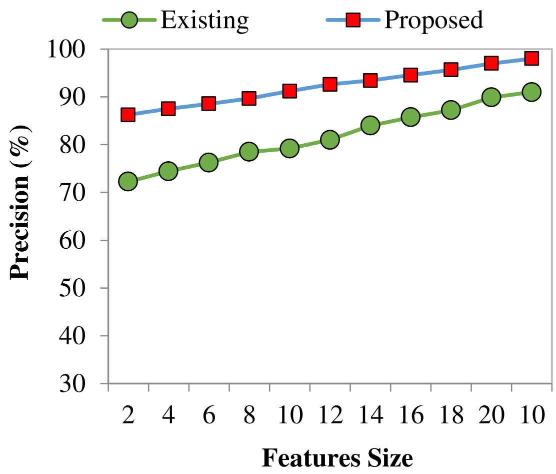


(b)

Fig.4. Accuracy (a). Sample Size, and (b). Feature Size

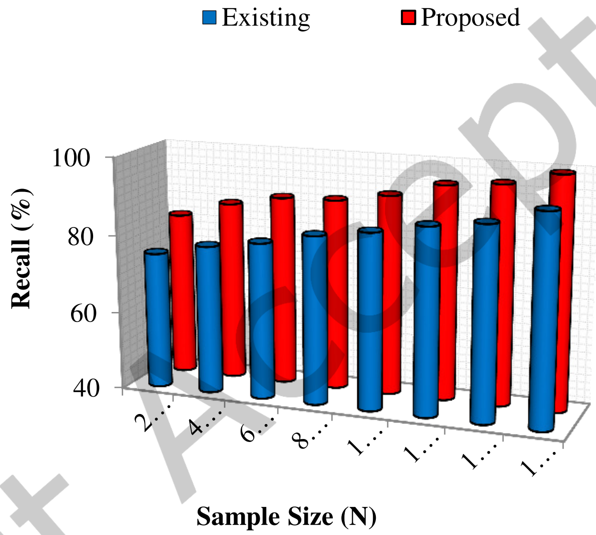


(a)

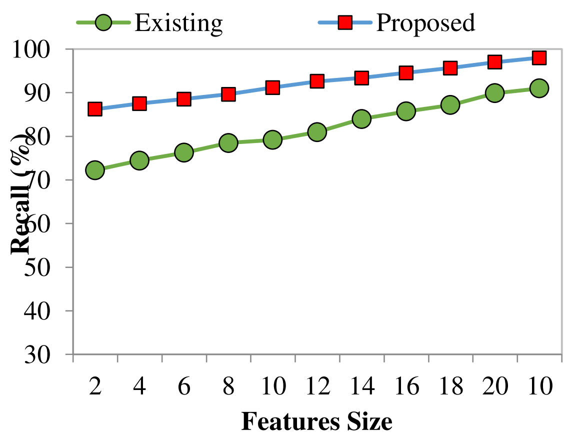


(b)

Fig.5. Precision (a). Sample Size, and (b). Feature Size

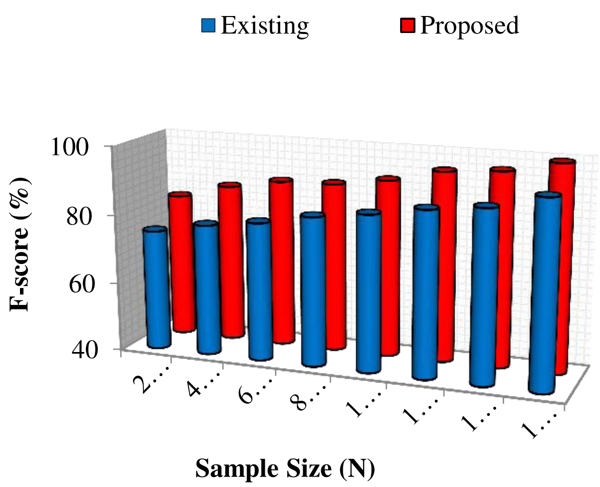


(a)

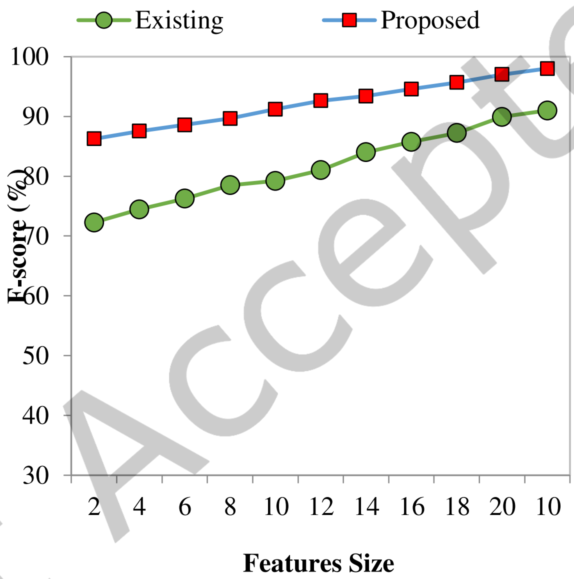


(b)

Fig.6. Recall (a). Sample Size, and (b). Feature Size



(a)



(b)

Fig.7. F-score (a). Sample Size, and (b). Feature Size

The observations from the experiments can be summarized as follows:

Data Augmentation Benefit: The model showed significantly lower loss when data augmentation techniques were applied. This suggests that data augmentation was effective in improving the accuracy of estimating acoustic features. By generating additional augmented data, the model had access to a larger and more diverse training set, allowing it to capture a wider range of acoustic variations.

Improved Generalization: In the case of data augmentation, there was a notable reduction in the gap between the training and validation losses compared to the case without augmentation. This indicates that the model's generalization performance was enhanced. The model consistently provided accurate estimation results not only for the training data it had seen but also for unseen data in the validation set. The narrower gap suggests that the model learned more robust and generalizable representations, resulting in better performance on unseen data.

These observations highlight the benefits of data augmentation in improving the accuracy of acoustic feature estimation and enhancing the model's ability to generalize to unseen data. By leveraging augmented data, the model gained a broader understanding of the acoustic variations, leading to more reliable and consistent estimation results.

5 CONCLUSION

In this study, we introduce GAN-TTS, an end-to-end multilingual TTS model that produces natural and human-like speech. Due to the limited availability of multilingual speech data for training, we employ a data augmentation approach to enhance the training corpus. One of the challenges in developing multilingual TTS models is accurately predicting speech in languages other than the target speaker's native language. To address this challenge, we incorporate a speaker regularization loss, allowing the model to learn speaker representations in a language-agnostic manner. This approach enables the model to handle multilingual synthesis effectively.

Furthermore, it is ensured that the model generates consistent phoneme durations regardless of speaker identity by substituting a zero vector for the speaker embedding during cross-lingual duration prediction. Additionally, leveraging DAT (Disentangled Acoustic Representations) and language embeddings, which are commonly used techniques in TTS models. GAN-TTS successfully produced high-quality audio samples in bilingual setups involving languages such as English, Korean, Japanese, and Mandarin Chinese. It achieves high speaker similarity in both cross-lingual and intra-lingual synthesis.

By demonstrating the effectiveness of GAN-TTS in languages beyond English, our research paves the way for future advancements in utilizing GAN-TTS for a wide range of languages, expanding its application and impact in the field of multilingual TTS synthesis.

Authors Contribution: Each author contributed equally in each part.

Data Availability Statement: The data used to support the findings of this study are available from the corresponding author upon request.

ETHICAL APPROVAL:

This article does not contain any studies with human participant and Animals performed by author.

CONFLICTS OF INTEREST

The authors are declared there are no conflicts of interest regarding the publication of this paper.

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